CROSSSAFE · RISK REGISTER

CrossSafe risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First risk register, preliminary hazard analysis and design FMEA (kit 1.9.1 upgrade); scores are the author's first estimate

CrossSafe risk register and hazard analysis

The top risks are that detection and visibility cannot be shown on paper (H3, F5), that the flash area near the button is above an analogue limit (H5, PR-016 At risk), and that a hot battery is a safety trade-off awaiting Amish (H2, O2). CrossSafe is a research and educational prototype at TRL 3; nothing here is a statement that the device is safe to put beside a road. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high), by the author, first pass.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	The pilot road authority will not accept a non-standard beacon	IA-21 and MUTCD status for this form not yet checked; local rules vary	No site; the project stays on paper	3	5	Design lead	Early review of the design against local rules (TRL 4); keep the pattern in firmware
P2	Cost stays above the value-engineering target	Saddles, plate, bracket and visors added for construction (CRS-DDR-003)	A school or market group cannot fund a crossing	4	3	Design lead	Quotes for the sign and enclosure; Amish decides O4
P3	Detection quality is not good enough (false or missed activations)	PIR plus radar behaviour in rain, wind and crowds is unknown	Drivers learn to ignore it, or pedestrians are not warned	3	4	Design lead	The button always works; site trial with manual counts (TRL 4)
P4	Hot-box battery ages quickly at the 55 C charge limit	CRS-DDR-005 raises the limit; shield factor is borrowed, not measured	Short battery life, early replacement	3	3	Design lead	Cell datasheet and ageing data (to assess); measure the shield
P5	Pole owner refuses a clamp-on kit	Utility poles are not the road authority's	No mounting point	3	3	Pilot partner	Own-post variant is costed ($437.00)
P6	Theft or vandalism takes the panel or battery	Panels and batteries have resale value	Lost device, cost	3	3	Local technician	Mount at 2.8 m or more; security fasteners (PR-014)
P7	The author's scoring of the options is wrong	Judgment from first-order numbers	Wrong concept continues	2	3	Design lead	Both A and C go to Amish; co-design tests UR-001 and UR-006 first
P8	Autonomy falls below 5 days in a cold region with an old battery and no heater	Charging inhibited below 0 C; end-of-life capacity 80 %; 70 % at -20 C (CAL-001 C7)	Beacon goes dark in a long cold spell	3	4	Design lead	18 Ah pack or a heater (O4); site climate check
P9	The existing pole cannot carry the 2,601 N m the kit adds	Thin or corroded pole; 76.1 x 3.2 mm pole has too little section modulus (CAL-001 G9)	Pole bends or falls	3	5	Pole owner	Owner verifies the pole; own-post variant; Amish decides O5

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Sign, bar or panel falls on a person	Wind overload, band slip, corrosion, vehicle strike	Head or crush injury	5	2	Post at 45 % of yield in a 40 m/s gust (PR-009); keyed saddles; bar at 2.1 m or higher; all fasteners security type	Medium: wind and strike loads are assumed; site check needed
H2	Lithium battery overheats, vents or burns	Hot box, overcharge, short circuit, crush	Fire, toxic gas, burns	5	2	LiFePO4 chemistry; BMS with low-temperature cut-off; 5 A fuse at the terminal; shield; battery at 2.8 m; charge limit set to the cell's rating (CRS-DDR-005)	Medium: charge at 55 C is a safety trade-off awaiting Amish (O2)
H3	Pedestrian steps out trusting a warning that is not working	Fault, dead battery, lost link, missed detection	Collision	5	3	Fault indicator and log (PR-012); timer-limited flashing; the sign still stands; wording that the device does not stop traffic	High until a trial: detection and visibility are not verifiable at TRL 3 (PR-002, PR-005)
H4	Fall from a ladder or lift during installation	Working at 4 m beside a road	Fall injury, traffic strike	4	2	Two-person crew; traffic control per local rules; shield fitted on the ground; install plan in the build plan	Medium
H5	Photosensitive reaction or discomfort from the flashing	Lit area of the bar near the button (7.7 msr at 1.5 m against a 6 msr analogue); five pulses a second in a sequence	Headache, in rare cases a seizure	4	2	Bar faces the road; the bar is above eye height (centre at 2.17 m) and faces the road; dimming at night; review by the road authority and a clinical-safety reviewer (to assess)	Open: PR-016 At risk; Amish decides O3
H6	Night glare or dazzle for drivers	Full-power amber at night	Reduced vision, a crash	4	2	Automatic dimming (PR-002); lens visors	Medium: dimming levels not set
H7	Electric shock or burns from the panel and wiring	20 W panel open circuit near 22 V; 12 V battery with 154 Wh	Burn from a short circuit	3	2	Low voltage only; fuse; covered cable entries; enclosure at 2.8 m	Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H8	Cuts and pinches while building the made parts	Sheet metal edges, drill, grinder	Cuts	2	3	Edges broken (kit model includes edge breaks); gloves and guards in BLD-001 safety stops	Low
H9	Radio interference with other devices	868 MHz LoRa duty cycle; US band differs	Interference, rule breach	2	2	Duty cycle 4.3 s against 36 s per hour (S8); US band to assess in REG	Low

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection (detection 1 easy to catch, 5 hard).

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Light bar and LED heads	Show the IA-21 pattern	Head stays dark	No warning although the sign says one	LED driver failure, open circuit	5	3	3	45	Open-LED detection in firmware (PR-012)
F2	Light bar and LED heads	Show the IA-21 pattern	Moisture inside the bar	LED corrosion, short circuit, reduced light	Failed seal at the visor or cover	4	3	4	48	Seal detail and a drain lip in the model; inspect yearly
F3	Controller and radio board	Start both sides within 0.5 s	Link lost	Only the local side warns	Antenna damage, blockage, interference	3	3	2	18	Local flashing without link (PR-012); fault log
F4	Controller and radio board	Limit flash time	Timer fails, flashing continues	Driver learns to ignore it; battery drained	Firmware fault	4	2	2	16	Hardware watchdog and a fixed upper limit (PR-012); to specify in firmware at TRL 4
F5	Presence radar and PIR	Detect a waiting pedestrian	Missed detection	Pedestrian is not warned	Occlusion, rain, a child below the beam	5	3	4	60	The button remains; trial counts (PR-005)
F6	Presence radar and PIR	Detect a waiting pedestrian	False activation	Warnings with nobody waiting; ignored	Wind, traffic, animals, a branch	3	4	3	36	PIR gates the radar; threshold set in trial
F7	Battery and BMS	Store energy for 5 days or more	Battery cut-off in cold or heat	No power, no warning	Charging below 0 C, or above the rated limit	4	3	3	36	BMS cut-off; 55 C rated cell (CRS-DDR-005); low-battery alarm
F8	Battery and BMS	Store energy safely	Thermal runaway	Fire in the enclosure	Overcharge, short, heat	5	1	3	15	LiFePO4, BMS, fuse, shield, 2.8 m mounting
F9	Solar panel and bracket	Charge the battery	Shaded or dirty panel	Less charge than 38 Wh/ day	Dust, bird droppings, leaves	3	4	2	24	Margin of 22 Wh/ day and break-even at 1.05 peak sun hours; clean yearly
F10	Enclosure and sun shield	Keep the battery below 60 C	Shield blocked or missing	Interior above 60 C	Fouled slots, lost shield	4	2	3	24	Shield part of the kit; check at inspection; measure at TRL 4
F11		Hold the sign and				4	3	3	36	

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
	Keyed pole saddles and bands	enclosure on the pole	Band slackens, sign twists	Sign points away from the road	Creep, corrosion, vandalism					Keyed saddles; two bands; check at inspection
F12	Post and footing	Carry wind loads	Footing too shallow for the soil	Post leans in a storm	Soil softer than assumed	5	2	3	30	Site check; deeper footing; the model uses 1,800 mm
F13	Push-button station	Start the warning; confirm by tone and light	Button jams or is out of reach	No activation, or the user cannot press it	Dirt, vandalism, height error	4	3	3	36	Passive detection still works; button at 1.05 m (PR-011); co-design
F14	Wiring and glands	Carry power and signals	Water enters at a gland	Short circuit, corrosion	Wrong gland, loose nut	4	3	3	36	Glands at the enclosure floor; drip loop; inspect

Highest RPN: F5 (60), F2 (48) and F1 (45). Failures that put a person at risk (F5, F8, F12) are kept on the list even when their RPN is lower, because their severity is 5.

Options B and C have their own failure modes, not analysed here beyond the STEM check; they would be analysed if Amish picks one (O1).